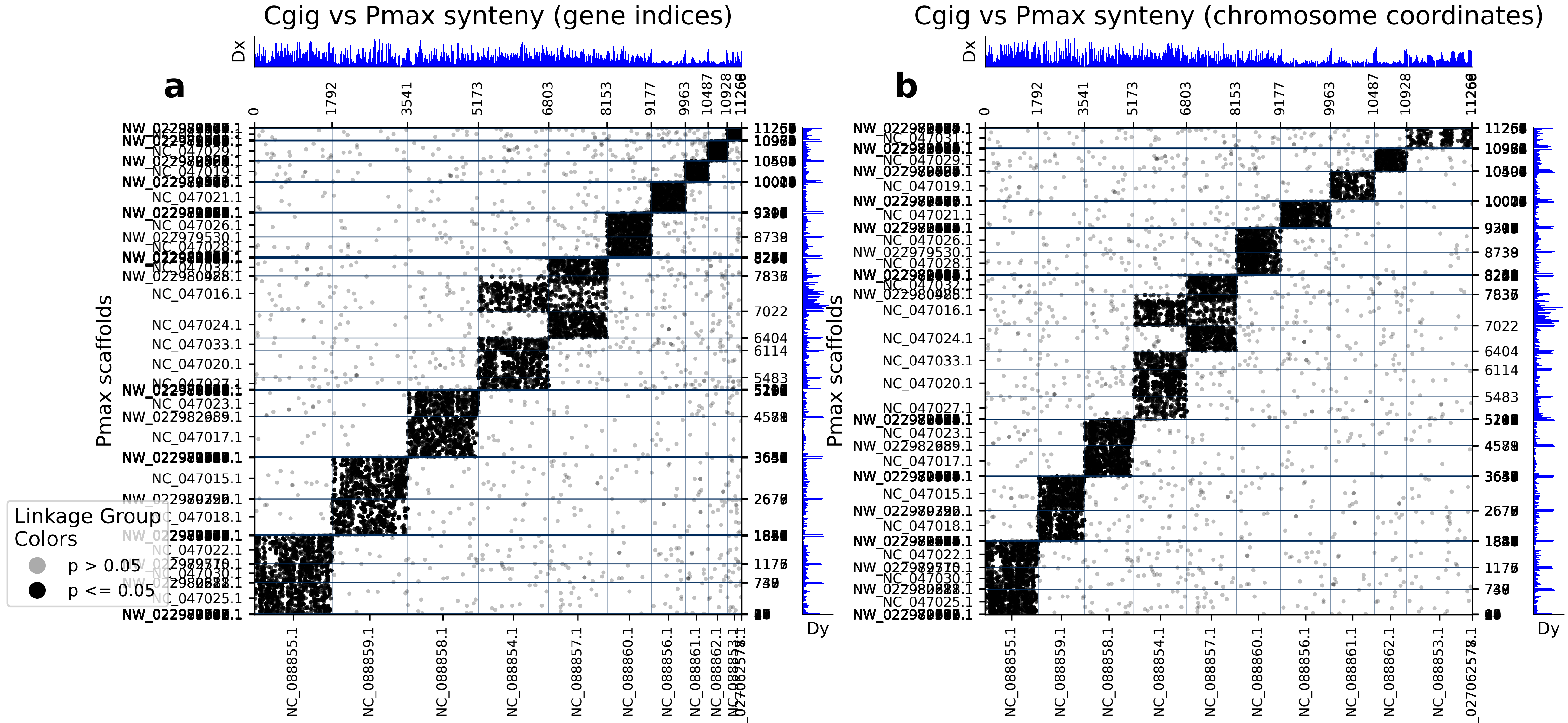
diamond: 11270 orthologs on 12 Cgig scaffolds and 249 Pmax scaffolds.



c

Cgig vs Pmax groups

Cgig scaffolds

d

Cgig vs Pmax FET, whole chroms.

	Cgig scaf	Pmax scaf	p (FET)	Count
0	NC_088858.1	NC_047017.1	0.00e+00	881
1	NC_088859.1	NC_047015.1	0.00e+00	870
2	NC_088859.1	NC_047018.1	0.00e+00	755
3	NC_088855.1	NC_047025.1	0.00e+00	652
4	NC_088856.1	NC_047021.1	0.00e+00	628
5	NC_088855.1	NC_047022.1	0.00e+00	579
6	NC_088854.1	NC_047020.1	0.00e+00	562
7	NC_088858.1	NC_047023.1	0.00e+00	557
8	NC_088857.1	NC_047024.1	0.00e+00	542
9	NC_088860.1	NC_047026.1	0.00e+00	493
10	NC_088854.1	NC_047016.1	2.50e-177	449
11	NC_088861.1	NC_047019.1	0.00e+00	403
12	NC_088860.1	NC_047028.1	0.00e+00	390
13	NC_088855.1	NC_047030.1	6.52e-235	371
14	NC_088862.1	NC_047029.1	0.00e+00	353
15	NC_088857.1	NC_047032.1	5.32e-273	352
16	NC_088857.1	NC_047016.1	9.95e-68	283
17	NC_088854.1	NC_047033.1	1.79e-153	241
18	NC_088853.1	NC_047031.1	8.75e-270	201
19	NC_088854.1	NC_047027.1	2.73e-105	196

e

Cgig scaffolds

Cgig vs Pmax FET, chrom. pieces

	Cgig scaf	Pmax scaf	p (FET)	Count
0	NC_088858.1 : 0-end	NC_047017.1 : 0-end	0.00e+00	881
1	NC_088859.1 : 0-end	NC_047015.1 : 0-end	0.00e+00	870
2	NC_088859.1 : 0-end	NC_047018.1 : 0-end	0.00e+00	755
3	NC_088855.1 : 0-end	NC_047025.1 : 0-end	0.00e+00	652
4	NC_088856.1 : 0-end	NC_047021.1 : 0-end	0.00e+00	628
5	NC_088855.1 : 0-end	NC_047022.1 : 0-end	0.00e+00	579
6	NC_088854.1 : 0-end	NC_047020.1 : 0-end	0.00e+00	562
7	NC_088858.1 : 0-end	NC_047023.1 : 0-end	0.00e+00	557
8	NC_088857.1 : 0-end	NC_047024.1 : 0-end	0.00e+00	542
9	NC_088860.1 : 0-end	NC_047026.1 : 0-end	0.00e+00	493
10	NC_088854.1 : 0-end	NC_047016.1 : 0-end	2.50e-177	449
11	NC_088861.1 : 0-end	NC_047019.1 : 0-end	0.00e+00	403
12	NC_088860.1 : 0-end	NC_047028.1 : 0-end	0.00e+00	390
13	NC_088855.1 : 0-end	NC_047030.1 : 0-end	6.52e-235	371
14	NC_088862.1 : 0-end	NC_047029.1 : 0-end	0.00e+00	353
15	NC_088857.1 : 0-end	NC_047032.1 : 0-end	5.32e-273	352
16	NC_088857.1 : 0-end	NC_047016.1 : 0-end	9.95e-68	283
17	NC_088854.1 : 0-end	NC_047033.1 : 0-end	1.79e-153	241
18	NC_088853.1 : 0-end	NC_047031.1 : 0-end	8.75e-270	201
19	NC_088854.1 : 0-end	NC_047027.1 : 0-end	2.73e-105	196

(a) depicts the chromosomes/scaffolds of Cgig (x-axis) plotted against the chromosomes/scaffolds of Pmax (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 11270 orthologs found between 12 Cgig scaffolds and 249 Pmax scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.